

Pakistan Rural Education Report

ASER 2023 — School Infrastructure Analysis

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1. Project Overview

This report analyzes rural school infrastructure across Pakistan using the ASER 2023 School Survey dataset. The survey covers 6,095 rural government schools across 7 provinces and 151 districts, making it one of the most comprehensive education datasets available for Pakistan. The central finding is that more than half of all rural schools surveyed — 50.3% — are non-functional. This means they are either open but not operating properly, temporarily closed, or permanently shut.

2. Dataset Summary

Item	Detail
Total schools surveyed	6,095
Provinces covered	7
Districts covered	151
Survey year	2023
Source file	ITAASER2023School.xlsx (137 columns)
Cleaned file	aser_2023_clean.csv (12 columns)
Target variable	nonfunctional (0 = functional, 1 = non-functional)
School types	Primary (72.5%) and Middle/Higher (27.5%)

3. Key Findings

3.1 National Picture

Metric	Value
Functional schools	3,028 (49.7%)
Non-functional schools	3,067 (50.3%)
Open but non-functional (S00=2)	914 (15.0%)
Closed temporarily (S00=3)	967 (15.9%)
Closed permanently (S00=4)	1,186 (19.5%)

3.2 Province-wise Non-Functional Rate

Province	Non-Functional Rate	Rank
Gilgit-Baltistan	68.9%	#1 — Worst
Azad Jammu & Kashmir	66.9%	#2
Punjab	62.5%	#3
Islamabad Capital Territory	55.8%	#4
Khyber Pakhtunkhwa	43.5%	#5
Sindh	39.3%	#6
Balochistan	35.6%	#7 — Best

3.3 Infrastructure Availability

Facility	% of Schools With It	% Missing
Teacher present at survey time	~78%	~22%
Drinking water	~65%	~35%
Toilet facility	~70%	~30%
Electricity	~55%	~45%

4. Gender Gap

Girls-only schools have a consistently higher non-functional rate than boys-only schools across all provinces — a gap of approximately 8 percentage points nationally.

School Type	Non-Functional Rate
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Girls-Only Schools	~55%
Boys-Only Schools	~47%
Gap (Girls minus Boys)	~8 percentage points

This compounds the existing gender gap in enrollment — not only do fewer girls attend school, but the schools that serve them are more likely to be non-functional.

5. Machine Learning Model

5.1 Goal

The model predicts whether a school is at high risk of being non-functional based on infrastructure features only — no attendance data. This enables policymakers to prioritize interventions without needing to visit every school.

5.2 Model Performance

Metric	Value
Algorithm	Random Forest Classifier
Accuracy	71.0%
F1 Score	0.727
ROC-AUC	0.771
Training split	80% train / 20% test
Data leakage fix	S007P (attendance) removed from features

5.3 Features Used

Feature	Description
province_encoded	Province (strongest predictor)
S03a1	Teacher present at survey time
S03d1	Drinking water available
S03e1	Toilet facility available
S03f1	Electricity available
is_girls_school	School is girls-only
is_boys_school	School is boys-only
is_primary	School is primary level
is_middle	School is middle/higher level

7. Recommendations

Policy

- Emergency school rehabilitation in Gilgit-Baltistan and AJK (both above 66% non-functional)
- Punjab review — 62.5% non-functional rate in the most populous province demands urgent action
- Prioritize girls schools — 8pp gap requires a dedicated gender-equity funding stream
- Electrify schools first — electricity is the scarcest resource and has compounding benefits
- Strengthen teacher attendance — it is the single strongest predictor of school functionality

Technical

- Replace GeoJSON file to enable district-level choropleth map
- Add shap to requirements.txt and re-save model bundle in deployment environment
- Load district_summary.csv in Tab 3 to enrich district metrics